

Emergent Overlays: Autonomous Adaptation for Efficient Broadcasting in Heterogeneous Mobile Ad-Hoc Networks

Abstract—Broadcast is a fundamental operation in Mobile Ad-Hoc Networks (MANETs). A large variety of broadcast protocols have been proposed to deal with network dynamics, maximize coverage and minimize the impact of collisions. These protocols follow either an unstructured approach based on the control of a flooding process or a more structured approach involving the creation of a dissemination overlay. The selection of the appropriate broadcast protocol for a MANET is a difficult task. Each protocol is better adapted to particular configurations of nodes, depending in particular on their density over space or their mobility. For heterogeneous node configurations, a single protocol is only adequate for some regions of the network offering poor performance elsewhere. This paper presents a novel adaptive approach to broadcast in heterogeneous MANETs. This approach allows MANET nodes to pick dynamically the most appropriate broadcast protocols for different regions of the network. We perform simulations and evaluate our approach using a mobility model that mimics how humans move to zones of interest, where overlays of nodes emerge as a result of applying our adaptive solution. Our evaluation results show that our adaptive approach leads to better performance and lower network consumption.

Raziel ► **ATM same abstract than before**◄

I. INTRODUCTION

Etienne ► *proposed introduction outline*

- MANETs are important due to advances in the IoT and the fact that not all systems will benefit from a fixed infrastructure (too costly, too seldom useful, etc.) [?]
- broadcast fundamental operation in MANETs, including for routing purposes, aggregation etc. [?], [?] - multiple protocols have been proposed for broadcast in MANETs
- broadcast in MANETs happens by the retransmission of a broadcast message by nodes until it reaches all nodes in the system. The objectives is to reach full coverage whilst spending as little per-node and overall energy as possible
- According to the survey by Ruiz and Bouvry [?], protocols can be classified in two main categories: protocols that use pre-established overlays, and those that do not use an established topology. This is similar to the typical unstructured vs structured overlays classification in P2P systems. Give a few details about the way unstructured approaches are based on controlled flooding (and therefore only use information available at the time of the propagation to decide on forwarding decisions) while approaches based on overlays can use information collected a priori and structure established also a priori.
- previous studies [?], [?] have shown that the performance and costs of each broadcast approach was intimately linked to environmental aspects of the MANET. In particular, density and mobility are key factors. The intuition is the denser the network is, the better it is to use an overlay to avoid collisions and speed up transmission (in less dense networks this is

not as helpful and may lead to partitions if not up-to-date); and, conversely, the more mobility there is, the less useful an overlay is since overlays constructed in such context quickly become deprecated, posing risks of having partitions or bad coverage.

- *choosing the appropriate protocol prior to the deployment of a MANET is difficult for a homogeneous network where the density and mobility patterns are not known a priori*
- *several MANETs are heterogeneous [?], [?], [?] and the density and mobility in different parts of the system will differ (find some convincing real life use cases to back this up, like the festival scenario)*
- *this makes the problem even more complex as the best algorithm for a part of the network may not be the best for another part.*
- *In this work, we are motivated by the notion that no one size fits all, and we are therefore interested in making broadcasting more adaptive and consequently more efficient and less costly for heterogeneous MANETs.*
- *our approach relies on a combination of interoperability and adaptation: (1) we allow different parts of the network to use different broadcast algorithms and ensure that coverage guarantees are maintained; (2) we propose dynamic adaptation mechanisms that allow nodes to dynamically select appropriate protocols based on the observation of the nodes deployment context. Our adaptation mechanisms range from simple approaches for a single node to coordinated approaches involving multiple nodes.*
- *We evaluate our approach using (include here highlights from the evaluation)*
- *Our contributions are (recap contributions)*
- *paper outline*

◄

II. EXPERIMENTAL ANALYSIS

Raziel ► **Experimental analysis: proposed outline.**◄

- Justification: MANETs fit well in such realistic scenarios where potentially there is no infrastructure.
- *TODO: Find a wireless topology (from realistic trace, protest or concert?) to emphasize the characterization of PoI*
- we characterize a PoI in zones according to the acceleration of nodes
 - zone Z_{sparse} with low density of nodes and high acceleration: where the Local Clustering Coefficient (LCC) of nodes is closer to 0
 - zone Z_{dense} with high density of nodes and low acceleration: with LCC of nodes closer to 1

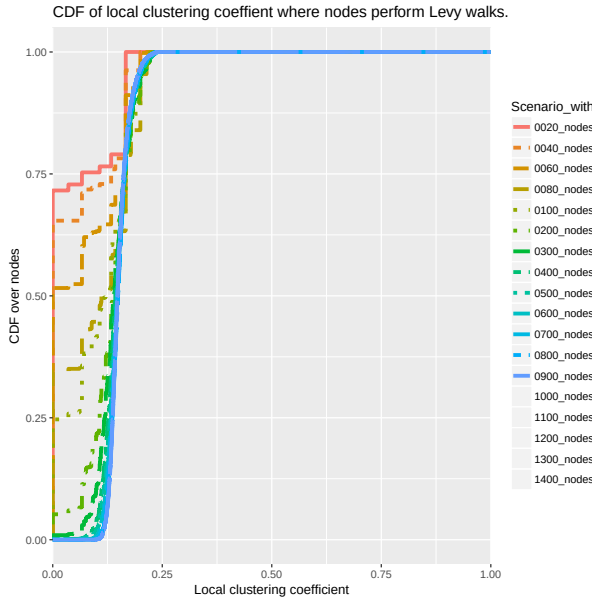


Fig. 1.

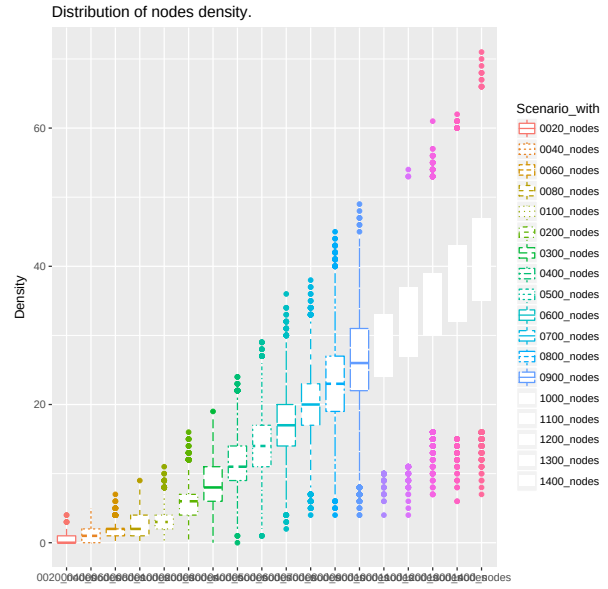


Fig. 3.

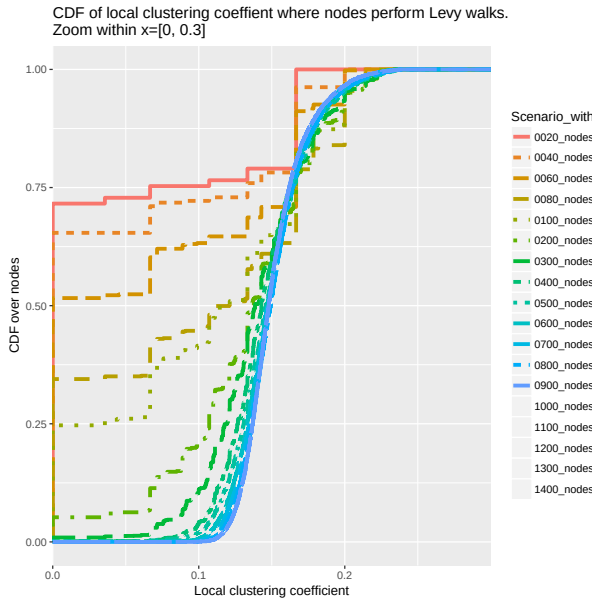


Fig. 2.

- Broadcast protocols in MANETs rely on the information available on each node to perform forwarding decisions (GPS coordinates, collisions in MAC, etc.)
- We use the acceleration of nodes as well as LCC (both local information) in broadcast algorithms for MANETs where nodes perform (truncated) Levy walks
 - let $z(l, a) = c(l) - n(a)$ be a function to decide if a node belongs to Z_{dense} or Z_{sparse}
 - $c(l) \in [0, 1]$ is the LCC
 - $n(a) \in [0, 1]$ is a normalization of the node acceleration.

TODO: find out trigonometric identity to map nodes acceleration within $[0, 1]$

– function $z \in [-1, 1]$

- * the more positive f the higher the probability that a node is at Z_{dense}
- * the more negative f , the higher the probability that a node is at Z_{sparse}
- * l is the list of two-hop neighbors; result of exchange of control messages

- Our policy of adaptation as well as the interoperability mechanism (between broadcast algorithms) rely on 2 factors: LCC and nodes acceleration. *TODO: highlight that no external entity is needed. Why LCC and acceleration.* both factors adapt themselves as needed. Here, getting LLC requires a periodic exchange of control messages; price to pay even in broadcast algorithms that do not rely in overlays. On the other hand, such exchange of beacons take place when a source node decides to disseminate a will of change (look at the algorithm of our basic policy of adaptation).

Raziel ► *Take into account the acceleration of nodes seems a natural choice to determine the zone where nodes move. On the other hand, for the local clustering coefficient we do require to analyze how this factor behaves in a wireless topology where nodes perform Levy walks.* ◀

- Ideally we should use real traces to study the LCC. **TODO: look for such traces**
- We do count with preliminary results to observe how LCC evolves in dynamic scenarios, this is shown in Figure 1, Figure 2, Figure 3; here there is a description of such experiments:
 - on each experiment there is a fixed number of nodes; such number varies from 20 to 1400

- communication area of $10m^2$ remains the same in all experiments
- for an experiment E_i a mobility trace T_i was created where every node performs Levy walks every second with a random velocity between 0m/s to 1m/s; total time on each trace is 30s
- transmission range of nodes remains the same for all experiments

III. ADAPTATION POLICIES

Etienne ► *the policies need better framing: we must present first what is the control/feedback loop and say we will present a few variants. We must describe clearly what is the problem first. We could separate between the data available to the nodes. I insist that having a table linking the currently-running protocol and the available data would be very useful, as it drives the type of adaptation policy we can have.*◀

Etienne ► *the proposed strategies (algorithms) integrate the threshold-based rules directly in the algorithm. It would be clearer to put them separately as predicates provided to the policies, and that can be reused across multiple decision-making strategies (e.g. the simple local one vs the complex collective one).*◀

Etienne ► *when using adaptation policies, it is necessary to use hysteresis to avoid constant oscillations between two protocols. We can also include a grace period where a node keeps its protocol for a time before being allowed to switch again.*◀ **Raziel** ► *proposal of simple adaptation policy*◀

- 1) all nodes are running A_i
- 2) source node forward message W_{msg} (wiling to change)
 - this action may take place when collisions increase (getting locally from MAC layer) or when there is a peak of energy consumption?
- 3) reception of message W_{msg}
 - a) forward W_{msg}
 - b) start sending control messages to eventually compute the local clustering coefficient (LCC) ($c(l)$)
 - c) log velocity V_0
- 4) all nodes in the communication area receive W_{msg} based in the way A_i works
- 5) when timer T_{will} (see 2.a) expires at the source node
 - a) log velocity V_1
 - b) get $n(a(V_0, V_1))$ where a is the acceleration
 - c) get LCC
 - d) stop sending beacons
 - e) IF $z(l, a) \approx 1.0$ THEN (node is at dense/static zone)
 - i) create message $C_{msg} = M(z(l, a), A_{cds})$ where A_{cds} refers to an algorithm that relies in an overlay
 - ii) forward C_{msg}
 - iii) switch to A_{cds}
 - f) IF $z(l, a) \approx -1.0$ DO steps e.[i - iii] with a non-overlay based algorithm identifier A_f (node is at sparse/dynamic zone)

6) Reception of C_{msg}

- a) IF $A_i \neq C_{msg}[A] \wedge z(l, a) - C_{msg}[z] \approx 0$ THEN
 - CASE: sender and receiver are within the same zone
 - forward message C_{msg} as it was received
 - start running new algorithm $C_{msg}[A]$ instead of A_i
- a) IF $A_i \neq C_{msg}[A] \wedge z(l, a) - C_{msg}[z] < 0$ THEN
 - CASE: first discovered border node
 - create message $C_{msg} = M(z(l, a), A_{flooding})$; **NOTE:** a node located at the border intends to reach other nodes at Z_{sparse}
 - forward C_{msg}
 - start running new algorithm $A_{flooding}$ which was taken from catalogs of algorithms
 - **NOTE:** when a border node forwards C_{msg} its identifier remains the same; meaning that receivers at Z_{dense} ignore such message

IV. OVERVIEW

Etienne ► *write a short intro to section*◀

Algorithm 1: Network layer API

```

1 send(m: message) : void           // send to all nodes in
  wireless range
2 receive(m: message) : void        // callback when a message
  is received

```

Algorithm 2: Broadcast protocol API

```

1 init(pl: payload) : message       // create a new message
2 emit(m: message) : void           // initiate a new broadcast
3 handle(m: message) : void         // handle an incoming
  broadcast

```

A. System model

Etienne ► *todo: add references to Algorithm 1 and Algorithm 2 in the text (if necessary)*◀

We generalize the architecture of MANET applications as three main layers. First, the network implements a wireless protocol of communication such as IEEE 802.11, where nodes share a common channel with Carrier Sense Multiple Access (CSMA) and collision detection capabilities. This network layer provides an API with two communication primitives:

- **send()**. This function disseminates a message in the transmission area $A_{emitter}$ **Etienne** ► *it is incorrect to use math mode for 'emitter' in $A_{emitter}$. If we actually need this notation (not sure it is used in the algorithms?) then we should use $A_{emitter}$ (see latex source)*◀ of the emitter
- **receive()**. This method allows the reception of messages for every node located within $A_{emitter}$

For the sake of simplicity, in the literature $A_{emitter}$ is depicted as a circle of radius T_x , which is the length of nodes' transmission range, centered at the position of the emitter.

Second, the Broadcast layer implements a dissemination protocol where a source node sends a message to the entire network by letting nodes act as relays and forward incoming

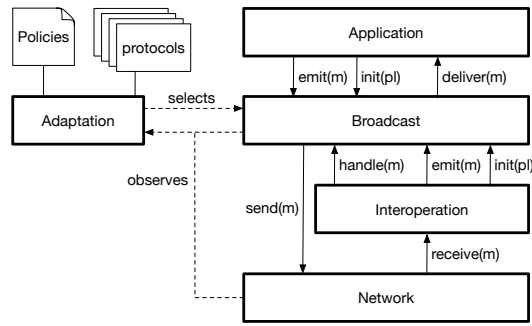


Fig. 4. Architecture of the system

messages. We present in form of API the main operations required by any broadcast protocol:

- **init()**: Nodes use this method to create a new message with the payload to disseminate. The metadata associated with this payload differs between protocols. For instance, a location-based technique may add the emitter's geographic position obtained from a GPS device to the metadata.
- **emit()**: Source nodes use this method to initiate a broadcast session.
- **handle()**: This method is triggered on every message reception and implements one broadcast protocol.

The third main layer is the MANET application per se and it is placed on the top of the Network and Broadcast layers. When the application requires to disseminate the information I , a new broadcast message is created by calling **init()** with I as argument. The dissemination of I starts with a call to **emit()**, its propagation to others nodes in the network is performed by **send()** and any message in the interest of the application will be delivered via **handle()**. **Etienne** ▶ *false, this is with deliver(). handle() is from the network to the bcst protocol*◀

The architecture of a MANET application is depicted on Figure 4 along with two more layers: Interoperation and Adaptation. These two layers extend an application with our adaptive approach in a network of nodes with heterogeneous configurations, by monitoring the dynamics of the network and choosing the best broadcast protocol, accordingly. **Etienne**

▶ *revise previous sentence, it may seem that the* ◀

B. Framework of adaptation

We are interested in designing an *adaptive* approach to broadcast in MANETs. Adaptation refers to the ability of the system to change its behavior based on the observation of its deployment environment and performance. In our target context, adaptation refers to the ability to dynamically select and configure the protocol used for broadcasting. It is important to note that this adaptation and protocol selection should be possible not only for the entire system but also, just for a part of it. Thus, we must guarantee interoperability among heterogeneous regions of nodes. **Etienne** ▶ *the following tries to provide more details than is needed here, yet is unclear. This section should provide the high-level of the separation between*

mechanism and policy and not detail how each work.◀ More concretely, every relay node between the source of a broadcast message and each destination node must perform a forward operation, such that there is any lost of messages when the emitter and receiver are not necessarily running the same protocol; our layer of interoperability deals with this regard.

1) *Interoperation layer:* **Etienne** ▶ *the content of this section is not in the right place. It is not the purpose of the overview section to present the details of how the interoperability and adaptation work. This is done in the two subsequent sections. Overall this section should be half a page, not more.*◀

We propose an interoperation mechanism between the Network and Broadcast layers, as shown in Figure 4. The basic idea is to broadcast a received message as a source node when the protocol of the emitter and the one from the receiver differ. **Vicent** ▶ *from?*◀ We assume that the protocol identifier is piggybacked on each broadcast message, note that this action does not modify the content of the message nor the metadata use by the protocols to their functioning.

Vicent ▶ *"on each broadcast message. Note that this action does not modify the message content, nor the metadata used by ..."*◀ This simple adaptation mechanism is depicted in Algorithm ?? . The function **receive()** (line ??) acts as a wrapper for the communication primitive with the same name at the Network Layer. **Vicent** ▶ *Here Layer is in capital but before wasn't. What do you prefer?*◀ When the protocol identifiers of the emitter and receiver nodes differ, a new message will be forward **Vicent** ▶ *forwarded?*◀ keeping the original payload (line ??) but using the receiver's protocol instead (line ??). This algorithm have a drawback, though a message with the same payload is forwarded twice when the protocol identifiers of two emitters differ. A more efficient adaptation mechanism is shown in Algorithm ?? where we use timers to delay.... **Raziel** ▶ *TODO:*

- *finish with the description of Algorithm ??*
- *argue that these simple and yet efficient mechanism guarantees coverage for any protocol and mention the proof in section 4*
- *create Broadcast super class to reduce the space used by all implementation of broadcast*
- *give an overall overview of the Adaptation layer saying that its detail explanation requires another section*

◀ **Etienne** ▶ *none of these things belong in section 3.*◀